Product Requirements Document (PRD)

Product Name: TaxShield Ledger (Micro-LLC Expense, Mileage & Payroll Tax Engine)
Target Market: Micro-Business LLCs (1–5 Employees / Contractors), Pre-revenue / Expense-heavy Holding & Operating Entities, Owner-Operators
Document Version: 1.0.0 (Planning Stage)
Status: In Review

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1. Executive Summary & Business Scenario

1.1 The Business Scenario

A micro-business LLC is established with **1 to 5 employees** (often the owner, family members, or part-time key operators). In early or holding phases, the LLC generates **minimal or zero gross revenue** while incurring legitimate ordinary and necessary business expenses (wages, travel, business mileage, tolls, parking, software, and administrative overhead).

Under IRS pass-through entity rules (Single-Member LLC via Schedule C / Form 1040, or Multi-Member LLC via Form 1065 K-1):

- **Net Operating Losses (NOL) and Business Expenses** pass directly through to the owner's personal tax return, offsetting active W-2 or investment income and legally lowering the owner’s net personal income tax liability.
- **Zero-Cost Operation Burden**: Paying
- 150/month for payroll software (Gusto, ADP) and
- 20/month for mileage apps (MileIQ) wipes out the tax efficiency for a micro entity. Owners instead use free tools (PayCheckCity, IRS withholding tables, myPATH, Zelle, manual logs), leading to disorganized records, severe calculation mistakes, missed filing deadlines, and extreme audit vulnerability.

1.2 The Core Problem

1. **Audit Risk on 5-Point Mileage & Travel**: The IRS routinely disallows vehicle deductions if contemporaneous 5-point logs (Date, Origin/Destination, Business Purpose, Start/End Odometer, Miles) and receipt trails for tolls/parking are not strictly kept.
2. **Payroll Compliance with
3. 52/yr), Federal 941, and year-end W-2/W-3 filing are legally mandatory. Manual calculations and portal copy-pasting are error-prone.

4. **Loss Flow-Through Verification**: The owner needs clear visibility into how LLC expenses and payroll costs accumulate toward their annual personal tax deductions (Schedule C / 1065 Pass-Through).

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2. Target Audience & Core Personas

Persona	Role	Key Goal	Pain Point
Owner-Operator	LLC Managing Member	Track legitimate expenses & mileage to maximize personal tax write-offs with $0 SaaS fees	Time-consuming manual spreadsheets, fear of IRS audit, confusing PA myPATH & local tax codes.
Family/Key Employee	W-2 (Part-time / Hourly)	Receive timely pay via Zelle and compliant pay stubs with correct withholdings	Needs official stubs and annual W-2 for personal tax filing.
Tax Preparer / CPA	External Advisor	Review year-end audit-proof logs and pre-aggregated box totals for Form 1040/1065	Receives messy shoebox receipts, missing odometer readings, and unreconciled payroll numbers.

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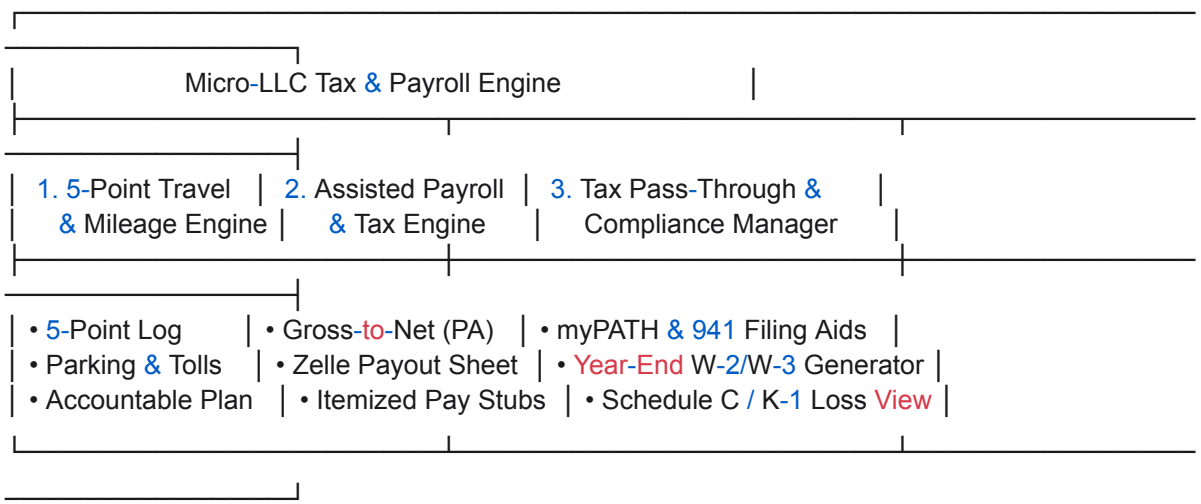
3. High-Level Product Goals & Principles

1. **IRS Audit-Proof by Design**: Strictly enforce mandatory IRS contemporaneous record-keeping requirements (cannot save a mileage trip without all 5 required points).
2. **Zero SaaS Cost / Assisted Manual Workflows**: Built specifically for manual Zelle/ACH payouts and manual portal submissions (PA myPATH, IRS EFTPS/Form 941, SSA W-2), eliminating all monthly payroll provider subscription fees.
3. **Pass-Through Tax Maximizer**: Provide real-time reporting showing total business deductions accumulated and their estimated offset value against the owner's personal tax bracket.
4. **Error-Free Tax Calculations**: Implement exact statutory engines (PA 3.07%, PA Local EIT by PSD, Federal Pub 15-T, FICA, LST) to replace ad-hoc manual calculators.

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4. Key Functional Modules & Use Cases

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Module 1: IRS 5-Point Mileage & Travel Expense Engine

Use Case 1.1: Log Contemporaneous 5-Point Business Trip

- **Trigger**: Owner or employee completes a business journey (client visit, equipment pickup, bank errand, administrative travel).
- **Required Fields (Enforced)**:
    1. **Date of Trip**
    2. **Origin & Destination Addresses / Locations**
    3. **Specific Business Purpose** (e.g., *"LLC Vendor negotiation & inventory pickup"* — flags vague inputs like *"meeting"*).
    4. **Starting Odometer Reading**
    5. **Ending Odometer Reading** (System calculates total miles).
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- **Output**: Calculates standard IRS deduction using the effective statutory rate (e.g., $0.67/mile).

Use Case 1.2: Attach Tolls and Parking Fees

- **Trigger**: Incurring associated travel costs during business travel.
- **Workflow**:
    - Add line items for Parking (garage, meter) and Tolls (PA Turnpike, bridge, highway).
    - Attach digital receipt image/PDF.
    - Choose payment source: *Owner Personal Out-of-Pocket* (to be reimbursed or deducted), *Company Card*, or *E-ZPass*.
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Use Case 1.3: Accountable Plan Reimbursement vs. Direct Tax Deduction

- **For W-2 Employees**: Approved mileage/tolls feed into the payroll engine as non-taxable expense reimbursements under an IRS-compliant Accountable Plan.
- **For Owner-Only / Schedule C**: Mileage and expenses aggregate directly into the annual business tax deduction ledger.

Module 2: Assisted Payroll & Gross-to-Net Engine (Zero-Cost Focus)

Use Case 2.1: Pay Period Computation & Exact Withholding

- **Trigger**: Bi-weekly, monthly, or ad-hoc pay run for 1–5 workers.
- **Computation**:
    - **Gross Wages**: Hours
    - Rate or Flat Salary.
    - **Federal Withholding**: IRS Pub 15-T bracket/percentage calculations based on 2020+ W-4 selections.
    - **FICA**: 6.2% Social Security + 1.45% Medicare (Employee & Employer portions).
    - **PA State Tax (SIT)**: Exact statutory **3.07%**.
    - **PA Local EIT**: Calculates higher of Resident PSD vs. Work PSD local tax rate (e.g., 1.00% to 3.8%).
    - **PA LST**: Prorated Local Services Tax deduction (e.g., $1.00/week up to $52/year).
    - **Non-Taxable Additions**: Accountable Plan mileage, tolls, and parking reimbursements added to Net Pay with **zero tax withholding**.
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Use Case 2.2: Zelle / Manual Bank Transfer Execution

- **Trigger**: Payroll run is calculated and locked.
- **Output**: Generates a consolidated **Zelle Transfer Matrix**:
    - Recipient Name, Phone/Email handle, Exact Net Amount in cents.
    - Checkbox & Reference Code input (e.g., `"Zelle Conf #ZEL-982341"`).
    - Marks transaction as reconciled once confirmed.
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Use Case 2.3: Compliant Pay Stub Generation

- **Output**: PDF & Web view showing clear separation of:
    - Gross Earnings
    - Itemized Taxes (Fed, FICA, PA-SIT, Local EIT, LST)

    - Non-Taxable Accountable Plan Reimbursements (Mileage, Parking, Tolls)
    - Net Take-Home Pay
    - YTD Accumulated Totals
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Module 3: Compliance Tasks, Agency Portal Filing Aids & Pass-Through Loss Tracker

Use Case 3.1: PA myPATH & IRS Form 941 Step-by-Step Filing Aids

- **Problem Solved**: Eliminates anxiety and arithmetic errors when logging into state/federal tax portals.
- **Workflow**:
    - Automatically creates scheduled compliance tasks:
        - **Federal**: Form 941 (Quarterly), Form 940 (Annual FUTA), EFTPS Tax Deposit deadines.
        - **PA State**: myPATH Form REV-1667 (Withholding Return), PA UC-2 (Unemployment Compensation).
        - **Local**: Local EIT Bureau returns (Berkheimer, Keystone, Jordan).
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    - **Portal Assistant View**: Side-by-side view showing the exact agency form box numbers and the exact values to copy-paste into the government web forms.
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Use Case 3.2: Year-End W-2 / W-3 Portal Package

- **Workflow**: Totals all annual payroll stubs and outputs pre-mapped box values for the SSA Business Services Online (BSO) portal and PA myPATH W-2 submission:
    - Box 1 (Taxable Wages), Box 2 (Federal Tax Withheld), Box 3 & 4 (Social Security), Box 5 & 6 (Medicare), Box 16 & 17 (PA Wages & 3.07% SIT), Box 18 & 19 (Local EIT).
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Use Case 3.3: Owner Pass-Through Tax Deduction Dashboard

- **Scenario**: LLC with $0 revenue, $15,000 in wages/taxes, $4,500 in 5-point mileage, and $1,200 in tolls/parking/supplies.
- **Output**:
    - Total Business Loss Generated: **$20,700**.

    - Pass-Through Benefit Simulator: Calculates estimated personal tax savings on Form 1040 Schedule C/E based on the owner's individual marginal tax bracket (e.g.,
    - 4,968 in personal tax reduction).
    - Exportable CPA Package (PDF/Excel) summarizing expense categories for instant tax return preparation.
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5. Non-Functional & Operational Requirements

1. **Data Accuracy**: All currency values stored in integer cents (`BIGINT`) to avoid floating-point rounding errors across tax agencies.
2. **Audit Readiness**: All mileage entries and pay records include creation timestamps, mileage start/end audit values, and document attachment URLs.
3. **Data Privacy & Security**: Employee SSNs and personal identification details are encrypted at rest with only last 4 digits exposed on the UI.
4. **Offline / Mobile Responsive**: Mileage and parking logs must be quickly accessible on mobile browsers for on-the-go entry right after a trip.
5. **Zero Vendor Dependency**: No recurring third-party API or payroll gateway dependencies required to operate.

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6. Success Metrics

- **Zero SaaS Cost**: 100% elimination of recurring payroll and mileage tracking subscription costs.
- **100% IRS Audit Compliance**: 0 missing mandatory fields on vehicle mileage logs.
- **Zero Calculation Discrepancies**: Exact alignment with PA Department of Revenue (3.07%) and IRS Pub 15-T wage bracket computations.
- **Time to File**: Under 5 minutes per quarter to complete manual portal filings using the side-by-side copy-paste Assistant View.